CMPE-202 Assignment 1: UML Class Diagram

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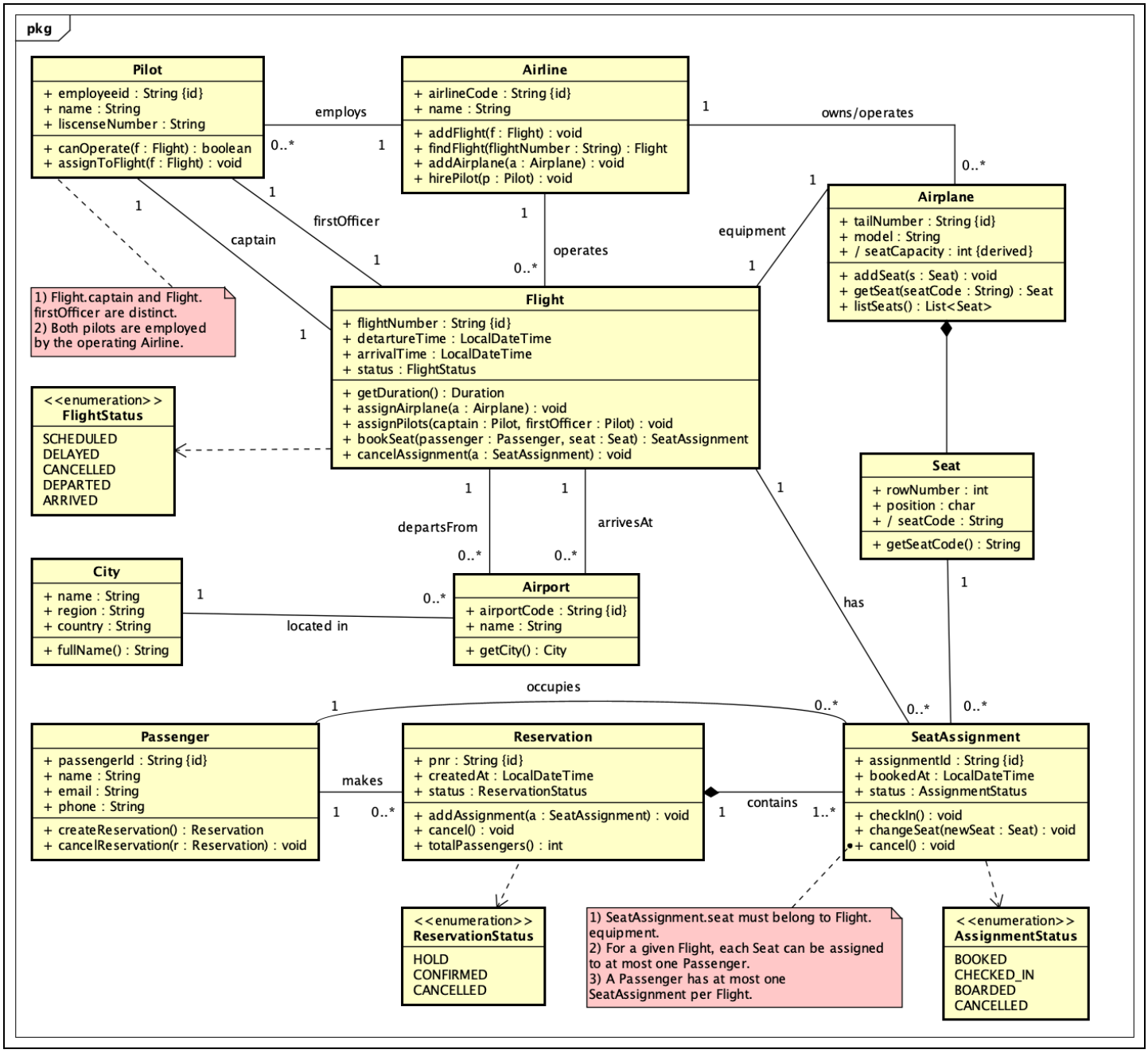
AirLine Reservation System UML Diagram



Java Class Declatations

```
public class Airline {
      public String airlineCode;
      public String name;

      public void addFlight(Flight f) {}
      public Flight findFlight(String flightNumber) {
            return null;
      }
      public void addAirplane(Airplane a) {     }
      public void hirePilot(Pilot p) {    }
}

public class Airplane {
      public String tailNumber;
      public String model;

      public void addSeat(Seat s) { }
      public Seat getSeat(String seatCode) {
            return null;
      }
      public List<Seat> listSeats() {
            return null;
      }
}

public class Airport {
      public String airportCode;
      public String name;

      public City getCity() {
            return null;
      }
}

public class City {
      public String name;
      public String region;
      public String country;

      public String fullName() {
            return null;
      }
}
```

```java
public class Flight {
      public String flightNumber;
      public LocalDateTime detartureTime;
      public LocalDateTime arrivalTime;
      public FlightStatus status;

      public Duration getDuration() { return null; }
      public void assignAirplane(Airplane a) { }
      public void assignPilots(Pilot captain, Pilot firstOfficer) { }
      public SeatAssignment bookSeat(Passenger passenger, Seat seat) {
            return null;
      }
      public void cancelAssignment(SeatAssignment a) { }
}

public class Passenger {
      public String passengerId;
      public String name;
      public String email;
      public String phone;

      public Reservation createReservation() {
            return null;
      }
      public void cancelReservation(Reservation r) { }
}

public class Seat {
      public int rowNumber;
      public char position;

      public String getSeatCode() {
            return null;
      }
}

public class SeatAssignment {
      public String assignmentId;
      public LocalDateTime bookedAt;
      public AssignmentStatus status;

      public void checkIn() { }
      public void changeSeat(Seat newSeat) { }
      public void cancel() {  }
}
```

```java
public class Pilot {
	public String employeeid;
	public String name;
	public String liscenseNumber;

	public boolean canOperate(Flight f) {
		return false;
	}
	public void assignToFlight(Flight f) {}
}

public class Reservation {
	public String pnr;
	public LocalDateTime createdAt;
	public ReservationStatus status;

	public void addAssignment(SeatAssignment a) { }
	public void cancel() {  }
	public int totalPassengers() {
		return 0;
	}
}

public enum FlightStatus {
	SCHEDULED,
	DELAYED,
	CANCELLED,
	DEPARTED,
	ARRIVED;
}

public enum AssignmentStatus {
	BOOKED,
	CHECKED_IN,
	BOARDED,
	CANCELLED;
}

public enum ReservationStatus {
	HOLD,
	CONFIRMED,
	CANCELLED;
}
```